

McKay, C.M. The Flinders University of South Australia, Bedford Park, S.A., Australia. Quantitative measurement of pteridines.

Preliminary investigations using paper chromatography techniques indicate that the two wild type stocks Canton-S and Loxton (isolated from a South Australian Population) plus the recessive mutations chocolate-2, purple, and sepia of *D. melanogaster* all differ chromatographically with

respect to the fluorescent eye pigments, pteridines.

Figure 1 shows that for the recessive mutations and wild types certain pigments: (1) may be totally lacking, (2) may be produced in greater quantity than either of the two wild types tested, (3) the quantity may be suppressed, or (4) new pigments may be formed which are not present in either of the two wild types tested.

Reference: Hadorn, E. and H. Mitchell, 1951 P.N.A.S. 37:650-665.

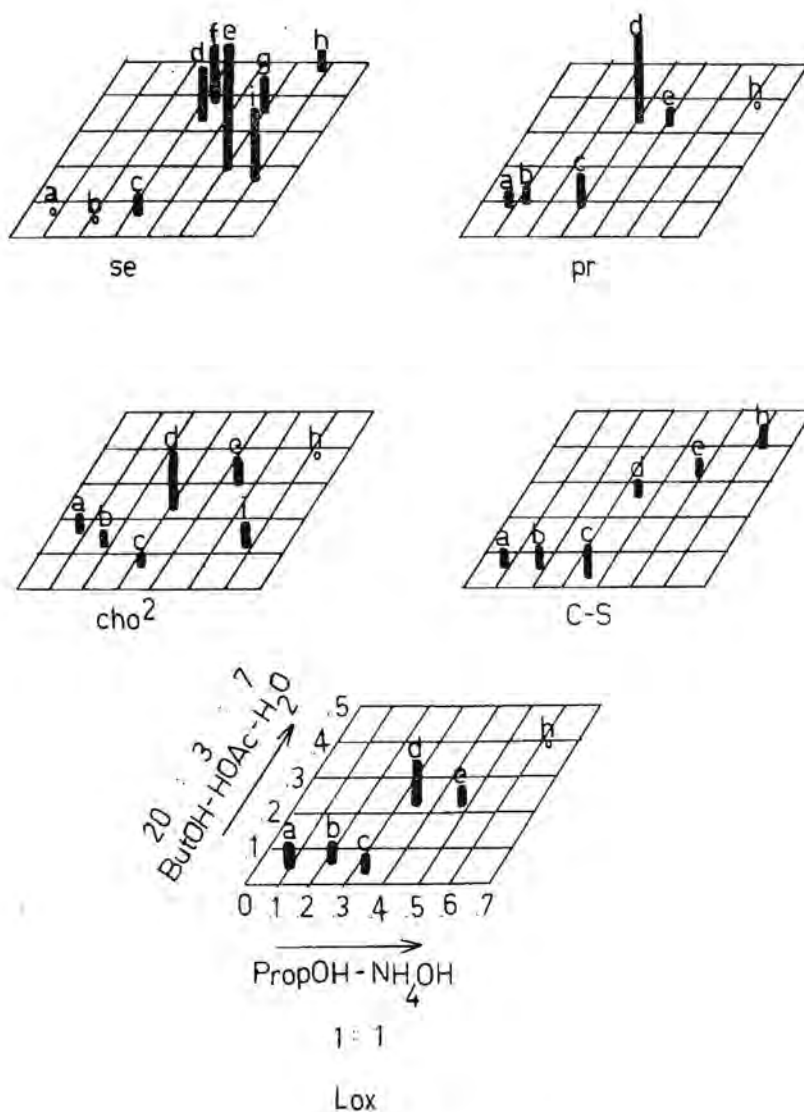


Figure 1. Three dimensional chromatograms. The heights of the columns represent the relative quantities of pteridines as measured by an Unicam SP. 800 Ultraviolet Spectrophotometer in two-dimensional chromatograms. Substances: a, drosopterin; b, isodrosopterin; c, isoxanthropterin I; d, isoxanthropterin II; e, xanthropterin I; f, xanthropterin II; g, sepiapterin; h, 2-amino-4-hydroxypterine; i, biopterin (Hadorn and Mitchell, 1951). The R_f values and solvent systems are the same for the five chromatograms as represented by the one for Loxton wild type. An open circle indicates the lack of the respective substance.